

Yatindra Rai

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Summary

2nd-year student and aspiring ML/AI Engineer skilled in designing production-ready full-stack AI applications using FastAPI, Next.js, and cloud deployment. Experienced in GenAI engineering including RAG, multi-agent orchestration, evaluation, and inference optimization with Groq. Research Intern at CMU focusing on EEG representation learning and deep learning for time-series signals.

Education

S.R.M Institute of Science and Technology

Chennai, Tamil Nadu

Bachelor of Technology in Computer Science and Engineering GPA: 9.38/10.00

May 2028

- Organization: SRM Placements (R&D Domain) – Domain Member
- Working with a team of 80 members conducting data mining and analysis of recruiting companies to support placement strategy

Skills

Languages: C++, Python

Backend: FastAPI, Python, RESTful APIs, Pytorch, Uvicorn

Frontend: Next.js, React.js, TailwindCSS

Artificial Intelligence: GenAI, LLM Fine-Tuning (LoRA), RAG Pipelines, Multi-Agent Systems, LLM Evaluation, EEG Representation Learning, TensorFlow, Analysis skills

LLM / Tools: Groq API, Ollama, Unsloth, HuggingFace, FAISS

DevOps / Tools: Docker, Git, GitHub, Render Deployment

Soft Skills: Strong problem-solving, quick learner, team collaborator with clear communication and analytical thinking.

Experience

Carnegie Mellon University

Dec 2025 – Present

Research Intern

Pennsylvania, USA

- Conducted research on EEG time-series modeling using deep learning architectures for brain signal analysis.
- Designed and evaluated hybrid sequence models combining FlashAttention and state-space architectures (Mamba) to efficiently model long high-frequency EEG signals.
- Implemented self-supervised learning and cross-subject evaluation pipelines to improve generalization across EEG datasets.
- Performed large-scale experiments and benchmarked model performance (accuracy, latency, memory) against baseline architectures such as GPT-2 and Transformer models.
- Developed preprocessing pipelines for multimodal brain data (EEG / fNIRS) including signal cleaning, feature extraction, and model-ready representations.

Projects

VeraciRAG – Ollama, RAG, Gen-AI, ChromaDB, Sentence Transformers

- [Project](#)
- Built VeraciRAG, a truth-verified multi-agent RAG system using Guardian → Generator → Evaluator agents with an automatic self-correction loop for higher answer reliability.
- Designed a fully local, zero-API-cost pipeline supporting privacy-friendly and air-gapped deployments while tracking confidence, sources, and correction metadata per query.
- Improved retrieval QA performance from 60% to 85% accuracy using BERT embeddings and FAISS for efficient similarity search.

NicheForge | Groq + Llama, LoRA Fine-Tuning, MLOps

- [Project](#)

- Developed NicheForge, a documentation-to-assistant system that automates synthetic dataset creation, LoRA fine-tuning of Llama models, and deployment via a chat interface.
- Automated synthetic Question and Answer data generation and LoRA fine-tuning on Llama models (Unsloth) for efficient specialization.
- Deployed a FastAPI + Next.js application with multi-backend inference fallback (Local, Groq, and Mock) to ensure reliable production behavior.

SynapseVideo — Multi-Modal Video Understanding Platform | *Multi-Modal RAG, Whisper, CLIP, Qdrant, FastAPI, Next.js*

- [Project](#)
- Built a multi-modal Retrieval-Augmented Generation (RAG) system enabling natural language search across video transcripts and visual frames using Whisper ASR, CLIP embeddings, and Qdrant vector database.
- Designed an async video processing pipeline using FastAPI, FFmpeg, and background tasks to extract audio, generate embeddings, and synchronize transcript segments with video frames via timestamp alignment.
- Integrated Groq Llama-3-70B with hybrid retrieval (vector search + SQL fallback) and developed a Next.js dashboard for video ingestion, processing status tracking, and real-time semantic search.

Achievements and certifications

Introduction to Modern AI

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional